

Uppsala University
Department of Information Technology

Literature Seminar

Written Preparation Report

Course: Data Engineering II

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1 Contextualization and Orchestration: Meaning, Importance, and Available Tools

2 Serverless Architecture, Digitalization of Society, and Big Data

3 Model Serving and CI/CD in Data Engineering

Role and Importance

Training a model is only the first step. To be useful, the model must run as a live service so other applications can send data and receive predictions — this is model serving. It manages model versions, routes requests, and scales with demand. CI/CD (Continuous Integration and Continuous Deployment) automates the steps between updating a model and deploying it to production. Because data changes over time, models need regular retraining and redeployment. Without CI/CD, this is slow and error-prone. Together, model serving and CI/CD let teams update models quickly, catch errors through automated testing, and keep predictions stable in production [1, 2].

Four Frameworks

1. **TensorFlow Serving (Model Serving):** An open-source tool by Google for deploying TensorFlow models. It serves predictions via REST or gRPC APIs and supports multiple model versions at once, which is useful for A/B testing. It is built for high-speed, high-load environments.
2. **MLflow Model Serving (Model Serving):** An open-source platform that works with any ML framework. A single command deploys a trained model as a REST API. It also provides a model registry to track versions and safely move models from testing to production.
3. **Jenkins (CI/CD):** One of the most widely used CI/CD tools. It can automatically retrain a model when new data arrives, run validation tests, and deploy passing models. Its many plugins connect it to cloud platforms, Docker, and ML frameworks.
4. **PerfCI (CI/CD):** A CI/CD framework designed specifically for ML pipelines. It automatically benchmarks model performance such as speed and accuracy. Each time a change is made, it flags regressions before the model reaches production. This makes it especially useful for teams that need to ensure model quality does not drop between updates.

4 Distributed vs. Federated Machine Learning

References

- [1] Bin Qian et al. “Orchestrating the Development Lifecycle of Machine Learning-based IoT Applications: A Taxonomy and Survey”. In: *ACM Computing Surveys* 54.1 (2020), pp. 1–48. DOI: [10.1145/3398020](https://doi.org/10.1145/3398020).
- [2] Joost Verbraeken et al. “A Survey on Distributed Machine Learning”. In: *ACM Computing Surveys* 53.2 (2020), pp. 1–33. DOI: [10.1145/3377454](https://doi.org/10.1145/3377454).